

Curvature-Bounded Black Holes and Compact Objects in CPTG

Finite Interior Structure, Mass-Spin Closure, and Strong-Field Admissibility

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Abstract

Curvature Polarization Transport Gravity (CPTG) provides a finite physical description of black holes while retaining the Kerr or Schwarzschild exterior wherever curvature polarization and curvature transport are negligible. Horizons remain admissible, but an infinite singular continuation is not accepted as the completed physical state. The black hole instead has a finite interior that joins the General Relativistic exterior cleanly and carries a resolved geometric response.

CPTG also changes the physical meaning of exterior mass. An angular-momentum-preserving mass-spin closure separates localized compact-source mass from the conserved exterior gravitational mass. A black hole can therefore produce the exterior gravity associated with a larger mass while containing less physical source mass, because part of the gravitational charge is supplied by geometry rather than an additional material component. On the registered compact branch, every nonextremal state has $M_c < M_\infty$, with the geometric share fixed by spin rather than object-specific tuning. Across the broader CPTG framework, the same source-versus-geometric-response principle replaces the cold-dark-matter sector of the Λ CDM model.

The compact state further determines an object-specific Structural Mode fingerprint, an organization shell, an inner trapping surface, finite-core clock behavior, and geometric frame dragging. The same finite solution class admits primordial black holes without stellar progenitors, and the same compact construction is validated from stellar black-hole benchmarks through the GWTC-5 binary-black-hole catalog and population posterior without object-specific compact retuning. CPTG therefore makes the black-hole interior calculable rather than singular: destruction of information at an infinite singular endpoint is excluded, Hawking evaporation must act on a finite source-geometry system, ordinary Kerr behavior is recovered where the CPTG response vanishes, and genuine ringdown departures must arise from CPTG strong-field dynamics rather than arbitrary exterior modification.

1 Introduction

Black-hole physics does not require CPTG to replace General Relativity. GR provides the exterior description of stationary black holes, orbital dynamics, gravitational lensing, horizon formation, and gravitational-wave propagation. CPTG retains that exterior wherever curvature polarization and curvature transport are negligible. Its strong-field requirement is that the completed physical compact continuation remain finite and match the ordinary GR exterior cleanly. Ordinary event or trapping horizons are therefore admissible; an infinite singular continuation is not accepted as a completed physical state. The strong-field transition is a smooth change in the relative importance of the geometric response sectors within one CPTG framework, not a switch between disconnected theories [25].

The CPTG compact construction uses a finite interior joined to the retained Schwarzschild or Kerr exterior at a defined matching boundary. A conditional mass–spin handoff distinguishes the physical compact-source mass from the exterior gravitational mass parameter, allowing the completed geometric response to contribute to the conserved exterior state. An exterior observer therefore measures the full gravitational charge rather than the localized source contribution alone: a compact object can gravitate as a larger mass while containing less physical source mass, with the difference carried by the solved geometry. Within the compact state, this is a source-versus-geometric-response distinction; across the broader CPTG framework, the same principle replaces the cold-dark-matter sector of the Λ CDM model with geometric response rather than additional material mass. The solved compact state also defines compact Structural Mode, an object-specific post-solution fingerprint of internal organization, together with a Structural-Mode shell, finite-core clock field, inner trapping surface, and geometric frame-dragging field. None of these downstream quantities is used as a fitting parameter to create the compact state.

The same compact equations apply to the stellar benchmarks M33 X-7 and LMC X-3 and to the declared 259-event GWTC-5 binary-black-hole sample without object-specific compact coefficients. In the gravitational-wave calculation, observation-level luminosity distance, detector-frame masses, and exterior spins are translated into the CPTG native coordinate before the compact handoff is applied. The resulting 520 component states preserve the exterior relativistic charges while resolving physical source mass, geometric mass-equivalent contribution, Structural Mode, interior clock behavior, trapping structure, and frame dragging. The separate selection-corrected GWTC population posterior is propagated through the same native-coordinate and compact-state maps.

The authority of the construction is deliberately layered. First, the finite spherical Misner–Sharp mass state closes the source, transport, and polarization channels. Second, a conditional mass–spin handoff preserves the exterior gravitational charge and total angular momentum. Third, Kerr-compatible rotating quantities—Structural Mode, trapping, clock, and ZAMO frame dragging—are evaluated downstream as post-solution readouts of that closed compact state. The third level is not a newly solved general axisymmetric interior field, and unique derivation of the complete rotating realization from the parent CPTG action for arbitrary strong-field geometry is not established. The finite-to-Kerr matching boundary is likewise not a universal polarization–transport activation radius: the location and width of the response crossover belong to the solved object rather than

to a universal horizon switch, curvature threshold, or mass scale. The present realization addresses the mass–spin compact sector; electromagnetic charge is not included, and this paper does not constitute a neutron-star equation-of-state calculation. This paper states that compact construction, its admissibility conditions, its internal geometric readouts, and its observational validation in a single strong-field description.

2 Parent CPTG Strong-Field Architecture

The compact sector begins from the same parent CPTG field organization used elsewhere in the theory. Ordinary physical stress–energy remains the material source, while curvature polarization and curvature transport are geometric response sectors:

$$G_{\mu\nu} + \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (1)$$

This distinction is important in the compact regime. The additional CPTG terms modify the geometric response to the physical source; they are not additional material sources placed on the right-hand side of Eq. (1).

The relationship to General Relativity is equally direct. Wherever the additional CPTG response becomes negligible,

$$\Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} \rightarrow 0, \quad (2)$$

Eq. (1) reduces to the Einstein equation with the same physical source. The ordinary Kerr or Schwarzschild exterior is therefore retained wherever Eq. (2) applies. An event or trapping horizon is not a boundary at which CPTG replaces GR, and the registered compact construction does not require such a switch. The finite interior must instead join the ordinary GR exterior cleanly at the compact matching boundary.

The strong-field problem is consequently one of finite continuation inside a common geometric framework. Curvature polarization and curvature transport can grow smoothly from negligible corrections into leading response sectors as the local strong-field state changes. Their crossover is not assigned a universal activation radius, horizon location, curvature threshold, or mass scale. The finite compact realization used in this paper supplies a concrete continuation and matching construction. Unique selection of that realization by the parent CPTG action, rather than membership in a broader admissible finite family, is not established.

3 Compact-Object Admissibility Rules

A CPTG compact solution must satisfy a small set of strong-field admissibility rules. These rules define the physical solution class and separate requirements of the parent theory from properties specific to the finite realization used in this paper.

3.1 Finite physical solution class

A physical CPTG black hole or compact remnant must possess a finite completed continuation. An infinite singular or unbounded-curvature continuation is not accepted as the completed physical state. This requirement does not discard the Einstein geometry outside the finite interior; it constrains how the compact solution is completed where the classical continuation would otherwise cease to remain physically admissible.

3.2 Ordinary horizons remain allowed

CPTG does not remove black-hole horizons. Event and trapping horizons remain admissible geometric features of the solved spacetime. A CPTG compact object may therefore possess the same ordinary Kerr or Schwarzschild exterior and horizon structure while differing only in its deeper finite continuation.

3.3 GR recovery and exterior matching are required

Where the polarization–transport response becomes negligible, the compact solution must recover the corresponding GR branch. For the registered compact construction used in this paper, the finite interior is required to join the ordinary exterior cleanly at its declared matching boundary. A finite interior that fails to match the required exterior is not an admissible completed compact solution.

3.4 The polarization–transport crossover is local and smooth

The polarization–transport crossover is not identified with the ordinary horizon and is not assigned to one universal radius. GR can remain the leading local description while the additional CPTG response grows continuously from negligible to dynamically important. The location and width of that crossover belong to the solved object and are not fixed by total mass class alone. The registered finite-to-Kerr matching coordinate used later in this paper is therefore a boundary of that realization, not a universal activation condition for the parent theory.

3.5 Physical compact-source mass and exterior gravitational mass are distinct

CPTG distinguishes the physical compact-source mass from the exterior gravitational mass parameter. The bookkeeping relation is

$$M_{\infty} = M_c + M_{\text{geom}}, \quad (3)$$

where M_c is the localized physical compact-source contribution of the stationary post-formation state and M_{geom} is the mass-equivalent contribution associated with the completed geometric response. M_c is therefore distinct from progenitor mass, pre-collapse

stellar mass, and the conserved exterior gravitational mass M_∞ ; the present construction does not require identifying it with one particular microscopic rest-mass definition. A positive M_{geom} therefore permits $M_c < M_\infty$: less physical compact-source mass can support the same exterior gravitational mass parameter because part of the exterior state is supplied by the solved geometry.

The finite compact realization supplies a conditional mass–spin handoff that computes this split from the exterior mass and spin state. The handoff is developed explicitly in the finite-interior section below. It defines the operational compact closure used in this paper; unique parent-action derivation of the same form for arbitrary rotating strong-field geometry is not established.

3.6 No universal stellar-scale lower source mass is imposed

The CPTG compact class is defined by the solved trapped geometry, finite continuation, and exterior match rather than by requiring a stellar amount of physical source mass. The compact framework therefore imposes no universal stellar-scale lower bound on M_c . Any lower bound arising from microscopic or semiclassical physics is separate from the strong-field construction stated here.

3.7 Scope of the finite realization

The registered minimum finite compact realization defines the interior construction used throughout this paper. It does not establish that every admissible compact object must possess one unique arbitrary-geometry interior profile. A broader finite family can share the same parent architecture, finite-continuation requirement, conservation structure, and GR exterior. Unique selection of the registered family by the parent CPTG action is not established.

3.8 Formation history and stationary compact structure are distinct

Collapse, fallback, radiation, angular-momentum transport, and merger dynamics determine how a compact state forms. The stationary compact solution describes the gravitational state after that dynamical history has selected its conserved exterior quantities. CPTG therefore treats formation history and stationary compact structure as related but distinct problems. The registered stationary construction does not by itself constitute a full dynamical collapse or merger calculation.

4 Registered Finite Compact Realization

The CPTG compact realization uses a minimum analytic finite continuation that preserves the ordinary Schwarzschild or Kerr exterior once the additional local CPTG response has completed. The construction is finite at the center, retains ordinary horizon structure,

and matches the exterior at a declared finite-to-GR handoff surface. It defines the interior used by the compact calculations in this paper. Unique selection of this profile by the parent CPTG action for every strong-field geometry is not established.

Write the normalized compact coordinate as

$$x = \frac{r}{r_m}, \quad 0 \leq x \leq 1, \quad (4)$$

where r_m is the finite-interior/GR matching radius. The registered minimum finite profile is

$$F(x) = \frac{1}{8} (63x^5 - 90x^7 + 35x^9). \quad (5)$$

At the outer edge of the finite continuation,

$$F(1) = 1, \quad F'(1) = 0, \quad F''(1) = 0. \quad (6)$$

These are matching conditions, not adjustable shape parameters. They make the normalized enclosed profile and its first two radial derivatives join the constant-mass exterior continuously at $x = 1$. The third radial derivative is not required to match. Within the retained minimum-degree analytic odd-polynomial realization, the conditions uniquely fix Eq. (5). That restricted uniqueness must not be confused with uniqueness over every finite compact geometry that could eventually be obtained from the parent theory. At the matching surface the local active polarization and transport generators have completed, while their integrated mass-equivalent contribution remains carried by the conserved exterior charge.

The central behavior of the registered minimum profile is also explicit:

$$F(x) = \frac{63}{8}x^5 + \mathcal{O}(x^7), \quad x \rightarrow 0. \quad (7)$$

The dimensionless radial carrier used later in the compact Structural-Mode construction therefore satisfies

$$U(x) \equiv \frac{F(x)}{x^2} = \frac{63}{8}x^3 + \mathcal{O}(x^5) \rightarrow 0. \quad (8)$$

Thus the retained minimum radial profile softens toward the center instead of developing a point-mass-like x^{-2} divergence, consistent with the finite, curvature-bounded compact solution class required by CPTG.

The registered boundary is a geometric handoff surface, not a universal polarization–transport activation radius. The local polarization and transport response can vary through the interior and need not switch on at r_m . Likewise, r_m is not an event horizon by definition and must not be conflated with any inner trapping surface or with the later Structural-Mode organization shell. The compact construction therefore preserves three logically different ideas: the ordinary exterior horizon structure, the finite-to-GR matching surface, and the internal distribution of the CPTG response.

For the retained compact state, the matching radius is

$$r_m = \frac{GM_\infty}{c^2}, \quad (9)$$

so the outer edge of the normalized finite realization is $x = 1$. The exterior mass M_∞ remains the conserved Schwarzschild/Kerr gravitational mass parameter seen outside the completed compact response. The finite interior does not replace that exterior charge; it resolves how that charge is assembled from the physical compact source and the completed geometric response.

4.1 Spherical source–transport–polarization mass organization

The finite compact state is first closed in spherical symmetry before any rotating handoff is applied. CPTG uses the Misner–Sharp mass as the covariant enclosed gravitational mass of that spherical state and defines the normalized interior coordinate

$$\mathcal{F}[g] = \frac{m_{\text{MS}}[g]}{M_\infty} = \frac{c^2 r}{2GM_\infty} (1 - D^a r D_a r). \quad (10)$$

The normalized state records the fraction of the final exterior gravitational charge enclosed at radius r . The physical source and completed geometric fractions are

$$\alpha_c = \frac{M_c}{M_\infty}, \quad f_{\text{geom}} = 1 - \alpha_c. \quad (11)$$

On this finite spherical state, the minimum no-extra-shape source–transport–polarization decomposition is

$$\mathcal{M}_c[g] = M_c \mathcal{F}, \quad (12)$$

$$\mathcal{M}_{\text{tr}}[g] = M_{\text{geom}} \mathcal{F}(1 - \mathcal{F}), \quad (13)$$

$$\mathcal{M}_{\text{pol}}[g] = M_{\text{geom}} \mathcal{F}^2. \quad (14)$$

These channels close identically:

$$\mathcal{M}_c + \mathcal{M}_{\text{tr}} + \mathcal{M}_{\text{pol}} = M_\infty \mathcal{F} = m_{\text{MS}}. \quad (15)$$

Their roles are distinct. The source channel follows the localized physical compact matter, the transport channel represents the finite interior redistribution active through the transition, and the polarization channel carries the geometric contribution as it accumulates into the completed exterior gravitational charge. For the registered minimum spherical realization, $\mathcal{F}$ is represented by the finite profile $F(x)$ of Eq. (5). At the outer handoff, $\mathcal{F} = F(1) = 1$, so

$$\mathcal{M}_c(r_m) = M_c, \quad \mathcal{M}_{\text{tr}}(r_m) = 0, \quad \mathcal{M}_{\text{pol}}(r_m) = M_{\text{geom}}. \quad (16)$$

Because Eq. (6) also gives $F'(1) = F''(1) = 0$, the local transport and polarization generators vanish at the handoff even though the completed integrated geometric contribution

remains in the conserved exterior mass M_∞ . This is the mechanism by which the finite interior can end in an ordinary GR exterior without requiring M_{geom} to disappear.

The authority of the next step is narrower: the spherical mass state above is already closed, while rotation is introduced through the conditional mass–spin boundary handoff and then through downstream Kerr-compatible structural readouts. The latter do not replace Eq. (14) with a separately fitted axisymmetric mass law and should not be interpreted as a completed general axisymmetric interior field solve.

5 Conditional Mass–Spin Handoff and Mass Interpretation

The compact mass accounting carries the already closed spherical source–transport–polarization state into a rotating exterior benchmark without changing its radial mass law. The handoff separates the physical compact-source mass from the exterior gravitational mass parameter while preserving the same total exterior charge. In the registered handoff,

$$M_\infty = M_c + M_{\text{geom}} = A_{\text{pol}} M_c, \quad (17)$$

where M_c is the localized physical compact-source contribution of the stationary post-formation state, M_{geom} is the completed geometric polarization mass-equivalent contribution, and $A_{\text{pol}} = M_\infty/M_c$ is the compact polarization factor. A positive M_{geom} therefore permits $M_c < M_\infty$ without changing the exterior gravitational charge. This is the boundary-level decomposition of the same conserved exterior state already organized by Eqs. (14)–(15), not the addition of unseen material.

Rotation closes the minimum handoff conditionally. Define the dimensionless exterior and compact-source spin parameters using the same conserved angular momentum J ,

$$\hat{a}_\infty = \frac{cJ}{GM_\infty^2}, \quad \hat{a}_c = \frac{cJ}{GM_c^2}. \quad (18)$$

Writing $q \equiv |\hat{a}_\infty|$, the retained Kerr exterior has the ordinary outer-horizon radius

$$r_+^{\text{Kerr}} = \frac{GM_\infty}{c^2} \left(1 + \sqrt{1 - q^2} \right) = r_m \left(1 + \sqrt{1 - q^2} \right). \quad (19)$$

Therefore

$$r_m < r_+^{\text{Kerr}} \quad (0 \leq q < 1), \quad r_m = r_+^{\text{Kerr}} \quad (q = 1). \quad (20)$$

The finite-to-Kerr handoff therefore lies inside the ordinary outer horizon throughout the subextremal branch and coincides with it only at exact extremality. The matching surface neither removes nor replaces the black-hole horizon.

The retained finite-to-Kerr handoff requires the same matching radius to be obtained from the exterior and compact-source descriptions,

$$r_m = \frac{GM_\infty}{c^2} = \frac{GM_c}{c^2} \left(1 + \sqrt{1 - \hat{a}_c^2} \right). \quad (21)$$

Using Eq. (17), this gives

$$A_{\text{pol}} = 1 + \sqrt{1 - \hat{a}_c^2}, \quad 1 \leq A_{\text{pol}} \leq 2. \quad (22)$$

Conservation of J independently gives

$$\hat{a}_\infty = \frac{\hat{a}_c}{A_{\text{pol}}^2}, \quad \hat{a}_c = A_{\text{pol}}^2 \hat{a}_\infty. \quad (23)$$

Eliminating $\hat{a}_c$ yields the compact closure used operationally from the exterior state,

$$\hat{a}_\infty^2 A_{\text{pol}}^3 + A_{\text{pol}} - 2 = 0. \quad (24)$$

For $0 \leq |\hat{a}_\infty| \leq 1$, the physical branch is the root satisfying $1 \leq A_{\text{pol}} \leq 2$. Once that root is selected, the compact-source and geometric contributions follow directly,

$$M_c = \frac{M_\infty}{A_{\text{pol}}}, \quad M_{\text{geom}} = M_\infty - M_c. \quad (25)$$

Equations (17)–(25) define the registered *minimum conditional mass–spin handoff* used throughout this paper. The handoff preserves the exterior gravitational charge and total angular momentum while separating the localized compact-source contribution from the completed geometric contribution. It is not a complete arbitrary-geometry rotating-interior field solution, and Eq. (24) is not established as a unique consequence of the parent CPTG action for every rotating compact geometry.

The distinction also clarifies the lower-mass question. CPTG does not impose a universal stellar-scale lower bound on M_c . The exterior mass inferred from an orbit, lensing field, or gravitational-wave signal is the exterior gravitational charge M_∞ ; the registered compact handoff allows a smaller physical compact-source mass when $M_{\text{geom}} > 0$. This does not by itself determine the formation history of a particular object. Progenitor mass, ejecta, fallback, radiation, neutrino losses, merger dynamics, and other source-history quantities remain part of the separate dynamical formation problem.

Lower-mass-gap systems such as GW190814 and GW230529 therefore remain useful observational targets [1, 2], but the existence of the compact handoff should not be turned into an object-specific classification without carrying the relevant observational state through the registered construction. The present result is the mass–spin closure itself; object-by-object applications require their own input authority and uncertainty propagation.

6 Stellar Compact Benchmarks

The registered mass–spin handoff can be tested on observed compact systems without changing the compact equations from object to object. Two deliberately contrasting stellar-mass black holes are useful for this purpose. M33 X-7 supplies a conventional high-spin reference branch, while LMC X-3 supplies a dynamically constrained low-spin

state. In both cases the observational mass and spin estimates are inputs to the conditional compact mapping; they are not re-fitted by CPTG, and the resulting compact quantities are downstream consequences of the same registered closure.

6.1 M33 X-7: conditional high-spin reference branch

M33 X-7 is retained as a conditional benchmark because published dynamical-mass and continuum-spin estimates exist, but the joint mass–spin state is model dependent. The original dynamical analysis reported

$$M_\infty = 15.65 \pm 1.45 M_\odot, \quad (26)$$

and the erratum-corrected conventional continuum branch gives

$$\hat{a}_\infty = 0.84 \pm 0.05 \quad (27)$$

[12, 13, 14]. A later spectroscopic analysis reported a lower black-hole mass near $11.4 M_\odot$ [18], while later continuum analysis showed substantial spin–plunging-region-stress degeneracy, including lower-spin solutions when intra-ISCO emission is retained [19]. The pair $15.65 M_\odot/0.84$ is therefore used here only as the retained conventional reference branch, not as a uniquely established model-independent joint state.

For the central reference pair,

$$M_\infty = 15.65 M_\odot, \quad \hat{a}_\infty = 0.84, \quad (28)$$

Eq. (24) gives

$$A_{\text{pol}} = 1.088925987. \quad (29)$$

The corresponding compact-source and completed geometric contributions are

$$M_c = 14.3719593 M_\odot, \quad (30)$$

$$M_{\text{geom}} = 1.2780407 M_\odot, \quad (31)$$

with

$$\frac{M_{\text{geom}}}{M_\infty} = 0.08166394, \quad \hat{a}_c = 0.996038237. \quad (32)$$

The registered matching radius is

$$r_m = \frac{GM_\infty}{c^2} = 23.1098807 \text{ km}. \quad (33)$$

Evaluating the source-side radius relation in Eq. (21) with M_c and $\hat{a}_c$ returns the same r_m to numerical precision. Thus the central branch assigns 8.1664% of the exterior gravitational mass to the completed geometric response, while the exterior gravitational mass is 8.8926% larger than the physical compact-source mass. At fixed $\hat{a}_\infty$, a different

adopted mass normalization rescales M_c , M_{geom} , and r_m while leaving the fractional split fixed by the conditional spin handoff; this is why the later lower mass estimate is not silently combined with the conventional high-spin value as though the two formed one measured joint posterior.

6.2 LMC X-3: low-spin comparison

LMC X-3 probes the opposite part of the same mass–spin relation. Its optical dynamical analysis gives

$$M_\infty = 6.98 \pm 0.56 M_\odot, \quad (34)$$

from the measured binary orbit and inclination [15]. Continuum fitting of more than one thousand thermal-state X-ray spectra gives

$$\hat{a}_\infty = 0.21^{+0.18}_{-0.22} \quad (35)$$

at 90% confidence [16]. A later IXPE spectropolarimetric analysis independently retained the low-spin interpretation, with contemporaneous spectroscopy again near $\hat{a}_\infty \simeq 0.2$ and a polarization-only 90% upper limit below 0.7 [17]. As with M33 X-7, the central values below define a conditional observed state rather than eliminating the model dependence of the spin measurement.

Using

$$M_\infty = 6.98 M_\odot, \quad \hat{a}_\infty = 0.21, \quad (36)$$

Eq. (24) gives

$$A_{\text{pol}} = 1.759699717. \quad (37)$$

The same compact closure then gives

$$M_c = 3.9665859 M_\odot, \quad (38)$$

$$M_{\text{geom}} = 3.0134141 M_\odot, \quad (39)$$

with

$$\frac{M_{\text{geom}}}{M_\infty} = 0.431721225, \quad \hat{a}_c = 0.650274050. \quad (40)$$

The matching radius is

$$r_m = \frac{GM_\infty}{c^2} = 10.3071544 \text{ km}. \quad (41)$$

The source-side relation again returns the same matching radius to numerical precision. For this low-spin central branch, 43.1721% of the exterior gravitational mass is supplied by the completed geometric response, and M_∞ is 75.9700% larger than the physical compact-source mass. The substantially different decomposition is therefore generated by the different observed mass–spin state, not by retuning the compact construction.

6.3 Direct comparison

The two central branches can be compared directly without introducing any new compact coefficient or object-specific prefactor.

Quantity	M33 X-7	LMC X-3
Exterior gravitational mass M_∞	$15.65 M_\odot$	$6.98 M_\odot$
Exterior spin $\hat{a}_\infty$	0.84	0.21
Polarization factor A_{pol}	1.088925987	1.759699717
Physical compact-source mass M_c	$14.3719593 M_\odot$	$3.9665859 M_\odot$
Geometric mass-equivalent M_{geom}	$1.2780407 M_\odot$	$3.0134141 M_\odot$
Geometric share M_{geom}/M_∞	8.1664%	43.1721%
Compact-source spin $\hat{a}_c$	0.996038237	0.650274050
Matching radius r_m	23.1098807 km	10.3071544 km

Table 1: Central conditional compact-state comparison for M33 X-7 and LMC X-3. The same registered mass–spin handoff is used for both objects; only the observational mass–spin state changes.

The comparison is important because it separates theory structure from object state. One compact relation accommodates both a high-spin branch with a relatively small geometric mass share and a low-spin branch with a much larger geometric share. The next section asks how these already solved states are organized internally. Structural Mode and the associated clock and rotation quantities are therefore evaluated only after the mass–spin closure has been completed; none is used to determine the compact state above.

7 Compact Structural Mode, Interior Time, and Geometric Rotation

The finite compact realization provides more information than the mass decomposition alone. Once the compact state is solved, CPTG can measure how the internal geometry is organized and can evaluate the local time and frame-dragging fields at that measured organization shell. These are downstream readouts of the solved state. Structural Mode is not a fit parameter, not an input to the compact closure, and not a prospective prediction assigned before the object is solved.

7.1 Post-solution compact Structural Mode

The registered Kerr-compatible rotating readout requires its own organization measure rather than importing the resolved-galaxy or spherical-cluster operator. Let

$$q = |\hat{a}_\infty|, \quad x = \frac{r}{r_m}, \quad (42)$$

and define the normalized equal-measure coordinate

$$\xi(x, q) = \left[\frac{x(x^2 + q^2)}{1 + q^2} \right]^{1/3}. \quad (43)$$

The radial profile entering the structural measurement is inherited from the registered finite realization,

$$U(x) = \frac{F(x)}{x^2}, \quad (44)$$

with structural invariant and weight

$$\mathcal{C}(x) = U(x)^2 + [U'(x)]^2 + 2 \left[\frac{U(x)}{x} \right]^2, \quad w(x) = |\mathcal{C}'(x)|. \quad (45)$$

The compact organization length and Structural Mode are then

$$\lambda_{N,\text{BH}} = r_m \frac{\int_0^1 \xi(x, q) w(x) dx}{\int_0^1 w(x) dx}, \quad (46)$$

$$N_{\text{BH}}(q) = \frac{r_m}{\lambda_{N,\text{BH}}} = \frac{\int_0^1 w(x) dx}{\int_0^1 \xi(x, q) w(x) dx}. \quad (47)$$

All explicit factors of mass, r_m , and c cancel from Eq. (47). Within this registered realization, the dimensionless fingerprint is therefore controlled by the solved exterior spin magnitude. It is nevertheless an object-specific post-solution measurement: a different solved compact geometry would be measured by the corresponding compact organization operator rather than forced to reproduce the present value. The spherical limit of the retained profile is

$$N_{\text{BH}}(0) = 1.912233171442, \quad (48)$$

and the registered branch decreases monotonically with increasing $|\hat{a}_\infty|$.

For the two stellar benchmarks already closed above, direct evaluation gives

$$N_{\text{BH}}^{\text{M33}} = 1.493871633444, \quad N_{\text{BH}}^{\text{LMC}} = 1.831967029641. \quad (49)$$

The difference is not introduced by a new compact coefficient. It follows from applying the same registered structural measurement to two already solved states with different exterior spin magnitudes.

7.2 The Structural-Mode organization shell

Structural Mode identifies a definite location inside the finite compact geometry rather than remaining only a scalar catalog label. The normalized organization-shell coordinate is recovered from

$$x_N(x_N^2 + q^2) = \frac{1 + q^2}{N_{\text{BH}}^3}, \quad r_N = x_N r_m. \quad (50)$$

For the retained central benchmark branches this gives

$$\text{M33 X-7: } x_N = 0.522687719562, \quad r_N = 12.0792508 \text{ km}, \quad (51)$$

$$\text{LMC X-3: } x_N = 0.527246370986, \quad r_N = 5.43440975 \text{ km}. \quad (52)$$

The similar normalized shell coordinates do not imply identical internal states. Their physical radii differ because each shell remains attached to its own solved compact scale r_m , while the mode itself records the organization of that object's solved geometry.

7.3 Structural Mode and the local structure of time

Once r_N is known, the registered Kerr-compatible rotating readout supplies a local clock field at that structurally selected location. Structural Mode does not replace the lapse field; it supplies the structural address at which the lapse is read. Using the same finite profile and $q = |\hat{a}_\infty|$, define

$$\delta(x, q) = x^2 - 2xF(x) + q^2, \quad (53)$$

$$\sigma(x, \theta, q) = x^2 + q^2 \cos^2 \theta, \quad (54)$$

$$\mathcal{A}_d(x, \theta, q) = (x^2 + q^2)^2 - q^2 \delta(x, q) \sin^2 \theta. \quad (55)$$

For the subextremal branch $0 \leq q < 1$, the physical inner trapping surface is the root $0 < x_- < 1$ of

$$\delta(x_-, q) = 0, \quad r_-^{\text{CPTG}} = x_- r_m. \quad (56)$$

At the exact extremal endpoint,

$$q = 1 : \quad \delta(1, 1) = 0, \quad x_- = 1. \quad (57)$$

The strict inequality $x_- < 1$ is therefore a subextremal statement. All benchmark and GWTC states evaluated numerically in this paper lie on that subextremal branch.

For the locally nonrotating, or ZAMO, geometry, the lapse is

$$\alpha(x, \theta; q) = \sqrt{\frac{\delta \sigma}{\mathcal{A}_d}}, \quad (58)$$

and its determinant-measure angular average is

$$\bar{\alpha}(x; q) = \frac{\int_0^\pi \alpha(x, \theta; q) \sigma(x, \theta, q) \sin \theta d\theta}{\int_0^\pi \sigma(x, \theta, q) \sin \theta d\theta}. \quad (59)$$

The Structural- N clock is therefore

$$T_{24,N}(q) = 24 \bar{\alpha}(x_N(q); q) \text{ h}. \quad (60)$$

It reports the proper-time accumulation at the organization shell during 24 source-local asymptotic exterior Kerr hours. Because x_N and $\bar{\alpha}_N$ are dimensionless functions of the solved spin state, this Structural- N clock is scale-free within the registered realization. For the two central benchmark states,

$$T_{24,N}^{\text{M33}} = 19.5877337 \text{ h}, \quad (61)$$

$$T_{24,N}^{\text{LMC}} = 13.1737826 \text{ h}. \quad (62)$$

This gives Structural Mode an additional compact-sector interpretation. It first measures where the solved geometry is organized, and that measured shell then carries a definite local clock readout. In this restricted sense, Structural Mode reveals the structure of time inside the registered finite black-hole realization: it does not predict time independently, but locates the solved shell at which the local time field is evaluated.

The organization-shell clock must not be confused with the clock arbitrarily close to the trapping surface. If

$$d = r_-^{\text{CPTG}} - r > 0, \quad (63)$$

then the simple-root behavior of δ gives the leading near-surface scaling

$$T_{24}(d) \propto \sqrt{\frac{d}{r_m}}. \quad (64)$$

The Structural- N clock is evaluated at the post-solution organization shell, whereas the near-trapping-surface clock is a separate scale-dependent local limit.

7.4 Geometric frame dragging at the organization shell

The same registered Kerr-compatible rotating readout also fixes the angular velocity of a zero-angular-momentum frame. The magnitude of the local geometric frame-dragging rate is

$$\omega(x, \theta; q) = \frac{c}{r_m} \frac{2q x F(x)}{\mathcal{A}_d(x, \theta, q)}. \quad (65)$$

This is a geometric ZAMO frame-dragging rate relative to source-local asymptotic exterior Kerr coordinate time. It is not a rigid-body rotation rate, a material angular velocity, or a fluid rotation law. Its determinant-measure angular average is

$$\bar{\omega}(x; q) = \frac{\int_0^\pi \omega(x, \theta; q) \sigma(x, \theta, q) \sin \theta d\theta}{\int_0^\pi \sigma(x, \theta, q) \sin \theta d\theta}. \quad (66)$$

Writing $\hat{\omega} = (r_m/c)\omega$, the organization-shell rate separates dimensionless geometry from physical scale,

$$\bar{\omega}_N = \frac{c}{r_m} \hat{\omega}_N(q). \quad (67)$$

For the two benchmark states, direct evaluation of the same determinant-weighted operator gives

$$\bar{\omega}_N^{\text{M33}} = 3683.246 \text{ rad s}^{-1}, \quad (68)$$

$$\bar{\omega}_N^{\text{LMC}} = 13248.618 \text{ rad s}^{-1}. \quad (69)$$

The larger physical rate for the lower-spin LMC X-3 benchmark is not a contradiction. The dimensionless rotational geometry is set by the solved spin state, while conversion to a physical angular rate also introduces the factor c/r_m . The smaller LMC X-3 matching radius therefore contributes directly to its larger physical rate; the angular rate should not be read as a one-parameter ordering by spin alone. Because q is the spin magnitude, these readouts give angular-speed magnitudes rather than a signed rotation direction.

7.5 Benchmark internal readouts

The central benchmark results can be displayed together without allowing any downstream quantity to flow backward into the mass-spin closure.

Quantity	M33 X-7	LMC X-3
Structural Mode N_{BH}	1.493871633444	1.831967029641
Organization coordinate x_N	0.522687719562	0.527246370986
Organization radius r_N	12.0792508 km	5.43440975 km
Inner trapping coordinate x_-	0.815809395077	0.597095316533
Structural- N clock $T_{24,N}$	19.5877337 h	13.1737826 h
Geometric frame dragging $\bar{\omega}_N$	3683.246 rad s ⁻¹	13248.618 rad s ⁻¹

Table 2: Post-solution internal readouts for the retained central M33 X-7 and LMC X-3 benchmark states. Structural Mode, organization shell, local clock, trapping coordinate, and geometric frame dragging are all evaluated only after the compact mass-spin state has closed.

Both benchmarks satisfy $x_N < x_- < 1$: the organization shell lies inside the inner trapping surface, while the finite-to-Kerr matching boundary remains at $x = 1$. The two-object calculation is not a population statement; its purpose here is to establish the meaning of the compact internal readouts before the catalog-wide validation is introduced separately.

8 GWTC-5 Catalog and Population Validation

The two stellar benchmarks establish the compact construction object by object. The GWTC-5 analysis provides the larger test: the same registered compact mass-spin closure, finite-core realization, post-solution Structural Mode, clock operator, and geometric frame-dragging operator are carried through the complete declared binary-black-hole analysis set without introducing event-specific compact coefficients or changing the structural rule between objects [20, 23, 24].

8.1 Event-level closure across the full declared sample

The event calculation begins from the released parameter-estimation information closest to observation. Luminosity distance, detector-frame component masses, and exterior spin magnitudes are used before source-frame reconstruction. The measured luminosity-distance posterior is held fixed and inverted through the registered CPTG affine luminosity observation coordinate to recover the native π -branch phase; externally reported source-frame masses and redshift are retained only as comparison and identity-audit coordinates [21, 20, 22].

The frozen source registry contains 259 astrophysical binary-black-hole events, 288 retained release-label realizations, and 260 unique within-event numerical streams after exact duplicate paired labels are removed only from catalog-level descriptive statistics. Genuinely distinct retained reconstructions remain separate. The compact solve therefore resolves 520 component states across 7,626,239 retained posterior draws. All 259 events close with no hard rejection under the declared fail-closed criteria [20].

Quantity	Audited result	Interpretation
BBH events	259	Full GWTC-5.0 BBH set
Release-label realizations	288	Provenance before deduplication
Unique numerical streams	260	Duplicate labels removed
Compact components	520	Two states per stream
Posterior draws	7,626,239	Full geometric propagation
Max. cubic residual	1.332×10^{-15}	Nonlinear closure
Max. mass residual	0	Exact mass closure
Max. detector/source residual	1.010×10^{-15}	Mass-coordinate audit

Table 3: Catalog-scale numerical closure of the registered CPTG compact construction. Residuals are numerical validation quantities, not observational uncertainties.

Structural Mode remains downstream of this solve. Across the retained component medians,

$$1.473147738840 \leq N_{\text{BH}} \leq 1.898389091913, \quad (70)$$

and the complete posterior audit preserves

$$x_N < x_- < 1, \quad \min(x_- - x_N) = 0.049408417286901. \quad (71)$$

No physical catalog-average N is assigned; each value remains attached to its own solved compact component and retained reconstruction. The corresponding component-median Structural- N clock spans approximately 11.8526–19.9856 h per 24 source-local exterior Kerr hours, while the determinant-averaged Structural- N frame-dragging rate spans 389.688–16330.834 rad s^{-1} . The clock and rotation quantities remain post-solution geometric readouts and are never used to retune the compact state [20].

8.2 Selection-corrected population push-forward

The population calculation is separate from the event-level structural analysis. It does not average the event Structural Modes, clocks, trapping radii, or frame-dragging rates and feed them back into the statistical model. Instead, the official GWTC-5.0 selection-corrected `default_mmax` hyperposterior is treated as an independent observational authority and passed through the same native-coordinate and compact-state maps [23, 24, 20].

The released population redshift is not treated as a CPTG-native observable. It is first decoded through the reporting convention carried by the GWTC product to recover the luminosity distance represented by that coordinate. The external values used by that decoder are therefore not CPTG inputs. The recovered luminosity distance is held fixed and the registered CPTG observation map is inverted for the native phase z_π . Only after that reconstruction is the population mass coordinate transformed. The exact scale is

$$s_M = \frac{1 + z_{\text{release}}}{1 + z_\pi}, \quad M_\infty = s_M M_{\text{source,release}}. \quad (72)$$

Across the 8200 hyperposterior-row medians, the released redshift-coordinate median is 1.400153, the recovered CPTG native phase is 1.528041, and the mass-coordinate factor has median 0.949412. The equivalent-standard H_0 -like quantity obtained after this solve is only a downstream reporting coordinate and does not enter Eq. (72), the compact masses, or the compact handoff [20].

All 8200 official hyperposterior rows are retained and each is propagated with 512 joint realizations,

$$8200 \times 512 = 4,198,400, \quad (73)$$

with no change to the event-side posteriors. Table 4 summarizes the resulting native-coordinate population transformation.

Per-hyperposterior-row median quantity	5th percentile	Median	95th percentile
Released component-1 mass coordinate	9.875	10.243	10.612
CPTG exterior component-1 M_∞	9.239	9.623	10.003
CPTG physical component-1 M_c	6.146	6.507	6.882
Released component-2 mass coordinate	8.007	8.507	8.947
CPTG exterior component-2 M_∞	7.865	8.407	8.879
CPTG physical component-2 M_c	4.560	4.837	5.087
Component-1 geometric share (%)	31.27	34.15	36.69
Component-2 geometric share (%)	42.54	44.04	45.19
Post-handoff reorder fraction (%)	17.19	22.46	27.54

Table 4: Native-coordinate push-forward of the official GWTC-5.0 selection-corrected population posterior. Mass entries are in $M_\odot$. Percentiles are taken across the 8200 hyperposterior-row medians. Released components retain the release ordering; CPTG components are ordered only after the compact handoff by physical compact-source mass.

At the median of the row medians, component 1 follows

$$10.243 M_\odot \longrightarrow 9.623 M_\odot \longrightarrow 6.507 M_\odot, \quad (74)$$

from released source coordinate, to CPTG exterior gravitational mass, to physical compact-source mass. Component 2 similarly gives

$$8.507 M_{\odot} \longrightarrow 8.407 M_{\odot} \longrightarrow 4.837 M_{\odot}. \quad (75)$$

The first shift is the native-coordinate reconstruction; the second is the compact mass–spin decomposition. The selection-corrected physical component-1 density correspondingly moves toward lower localized source mass: its pointwise median peaks in the $4\text{--}6 M_{\odot}$ bin rather than the $8\text{--}10 M_{\odot}$ bin of the released primary-mass coordinate. This does not remove the high-mass exterior systems from the input posterior. It changes the finite-state decomposition assigned to those exterior gravitational states, and the low-mass end retains the selection cautions of the released GWTC-5.0 BBH population analysis [23, 20].

The catalog and population calculations therefore test complementary aspects of the same construction. The event lane demonstrates repeated posterior-resolved closure of the compact state and its internal structural readouts. The population lane shows that the same compact relation remains operational when the independent selection-corrected posterior is translated through the CPTG native coordinate. Neither result is a detector-level native CPTG reanalysis of the strain likelihood; that remains a separate observational frontier.

9 Direct Strong-Field Consequences and Exclusions

The compact equations already decide several black-hole questions at the level of the stationary geometry. These consequences do not depend on a speculative formation scenario or on fitting an additional compact parameter. They follow from the finite continuation, the retained Kerr or Schwarzschild exterior, the conditional mass–spin closure, and the internal geometric ordering established above. Other questions, especially those requiring quantum evolution or a perturbation theory beyond the stationary background, remain separate from these direct consequences.

9.1 Finite continuation and the information problem

CPTG excludes an infinite singular continuation as the completed physical state. The classical singularity is therefore not available as a physical endpoint at which information, matter, or causal evolution can simply terminate. Any black-hole information-loss argument that requires destruction of degrees of freedom at an actual infinite-curvature singularity is incompatible with the CPTG compact solution. The classical GR singularity theorems remain statements about the assumptions under which they are derived, but CPTG changes the strong-field field content through the polarization and transport sectors and replaces the singular continuation by the finite geometry used here.

This does not, by itself, prove a complete quantum theory of information recovery. The entanglement and unitarity questions associated with Hawking radiation must be formulated on the finite CPTG background rather than on a spacetime whose physical

continuation ends at a singularity [7]. The definite result is narrower and important: an infinite singularity cannot serve as the physical sink of information in CPTG.

9.2 Hawking radiation and evaporation

CPTG does not remove the ordinary outer event horizon. For the same exterior mass and angular momentum, the stationary exterior is the corresponding Schwarzschild or Kerr geometry, including its outer-horizon structure. The usual semiclassical Hawking mechanism is therefore not eliminated merely because the deep interior is finite [6]. At the stationary-background level, CPTG provides no basis for replacing the standard exterior surface gravity by an arbitrary new value; any modification of Hawking emission must arise from the semiclassical CPTG evolution, backreaction, or the changing finite-state decomposition rather than from deleting the horizon.

The evaporation endpoint is a different question from the existence of Hawking emission. The stationary equations used here do not choose among complete evaporation, a finite remnant, or a transition into another compact state. They do impose one boundary on every such possibility: the physical evolution cannot terminate in the infinite singular continuation excluded by the CPTG strong-field solution class.

9.3 Exterior observables and gravitational ringdown

The exact recovery of the Kerr or Schwarzschild exterior places a strong restriction on where CPTG can differ from GR. Stationary observables determined entirely by the exterior background—including ordinary exterior orbital dynamics, lensing, and the leading shadow or photon-region geometry for the same conserved exterior charges—are not shifted simply because CPTG resolves the interior differently.

The same point constrains gravitational ringdown. Quasinormal modes are perturbative properties of a black-hole background and its boundary conditions [8]. The stationary CPTG background does not replace the exterior Kerr potential with a different metric. Therefore a ringdown deviation cannot be inserted arbitrarily as a consequence of the finite core. In the limit in which the linearized polarization and transport response is negligible in the exterior and the usual ingoing horizon condition is retained, the ordinary Kerr ringdown spectrum is recovered. Any genuine CPTG shift must instead come from the perturbation dynamics of the CPTG response sectors, the nonstationary merger-to-remnant transition, or a boundary condition that is itself derived from the finite theory.

This separation is observationally useful. Agreement of the dominant ringdown with GR is compatible with CPTG and is expected when the signal probes only the retained exterior. A measured departure is meaningful only if the CPTG perturbation equations predict it from the solved compact state rather than from an ad hoc modification of the exterior metric.

9.4 No generic reflective-surface echo

The registered CPTG compact object is not a horizonless material surface placed outside the would-be Kerr horizon. The ordinary horizon is retained and the finite-to-Kerr matching radius is an internal geometric handoff, not a reflective wall. Consequently, the standard echo mechanism in which radiation repeatedly reflects between an exterior potential barrier and a material surface replacing the horizon is not a generic prediction of this construction [9]. Finite interior structure alone is therefore insufficient to claim gravitational-wave echoes.

Secondary late-time structure is still possible if it follows from the CPTG perturbation equations or from nonstationary merger dynamics, but it must be derived from those dynamics. The stationary geometry rules out treating a reflective surface outside the horizon as an automatic consequence of CPTG.

9.5 Spin fixes the minimum source fraction

The conditional mass–spin handoff gives a direct and unusually restrictive prediction for the relation between the exterior gravitational mass and the localized physical source. Writing

$$q = |\hat{a}_\infty|, \quad q^2 A_{\text{pol}}^3 + A_{\text{pol}} - 2 = 0, \quad (76)$$

with the physical branch $1 \leq A_{\text{pol}} \leq 2$, implicit differentiation gives

$$\frac{dA_{\text{pol}}}{dq} = -\frac{2qA_{\text{pol}}^3}{1 + 3q^2A_{\text{pol}}^2} \leq 0. \quad (77)$$

Thus the geometric share decreases monotonically as the exterior spin magnitude approaches extremality. The endpoint limits are

$$q = 0 : \quad A_{\text{pol}} = 2, \quad q = 1 : \quad A_{\text{pol}} = 1, \quad (78)$$

so the registered handoff requires

$$\frac{1}{2} \leq \frac{M_c}{M_\infty} \leq 1, \quad 0 \leq \frac{M_{\text{geom}}}{M_\infty} \leq \frac{1}{2}. \quad (79)$$

For every nonextremal state on this branch, $M_c < M_\infty$. A black hole can therefore produce the exterior gravitational field associated with a larger mass while containing less localized physical source mass; the difference is carried by the solved geometry rather than by an additional material component. In the spherical limit the minimum conditional handoff assigns one half of the exterior gravitational mass to the physical compact source and one half to the completed geometric response, while the extremal limit approaches $M_c = M_\infty$. These are consequences of the registered mass–spin relation, not object-specific fit choices.

9.6 Internal ordering, clock behavior, and geometric rotation

The registered finite realization also makes definite statements inside the compact state. Structural Mode is solved only after the mass–spin state closes, and its organization shell remains distinct from both the inner trapping surface and the exterior matching boundary. For the subextremal branch $0 \leq q < 1$ represented by the benchmark and GWTC calculations,

$$x_N < x_- < 1. \quad (80)$$

The exact extremal endpoint instead satisfies $x_- = 1$, as shown in Eq. (57). Throughout the retained GWTC posterior audit the strict subextremal ordering is preserved, with minimum normalized separation $x_- - x_N = 0.049408417286901$. The organization shell therefore does not merge with or cross the inner trapping surface anywhere in the realized catalog states.

The local clock and frame-dragging fields have different trapping-surface limits. The ZAMO proper-time accumulation approaches zero with

$$T_{24}(d) \propto \sqrt{\frac{d}{r_m}}, \quad (81)$$

while the geometric frame-dragging field approaches a finite angle-independent trapping-surface rate in the registered rotating geometry. These are direct geometric predictions of the solved compact state and are independent of any assumed material rotation law [20].

Taken together, these consequences divide black-hole phenomenology into two classes. CPTG gives hard exclusions and predictions wherever the stationary finite geometry and matching relations are sufficient: no physical infinite singularity, no required horizon removal, no generic exterior reflective surface, a spin-dependent source/geometric mass split, fixed internal radial ordering, and definite clock and frame-dragging limits. Quantum unitarity, the detailed evaporation endpoint, tidal response, and any non-GR quasinormal-mode shift require additional dynamical or semiclassical equations, but those calculations must be performed on the finite background and obey the exterior and conservation structure already fixed here.

10 Allowed Compact Formation Channels

The stationary CPTG compact construction does not restrict finite compact objects to one astrophysical origin. Formation history determines how a system reaches its exterior mass, angular momentum, and matter state; the compact equations determine the finite stationary continuation associated with those boundary data. Every admissible realization must satisfy the same finite-curvature, conservation, trapping, and GR-recovery conditions.

10.1 Quiet stellar collapse and vanishing stars

NGC 6946-BH1 remains a major failed-supernova or vanishing-star candidate, while M31-2014-DS1 provides another massive-star disappearance case associated with black-hole formation [3, 4, 5]. These observations constrain source history but do not directly measure a CPTG interior. Their interpretation therefore requires progenitor, ejecta or fallback, radiative-loss, compact-source, and geometric-response accounting to remain distinct.

10.2 Ordinary stellar and merger formation

Black holes formed through core collapse, fallback, binary interaction, or compact-object mergers are compatible with the compact construction provided the final geometry satisfies the finite-solution and exterior-matching conditions. The stationary state is characterized by its solved mass, angular momentum, matter content, finite geometry, and exterior charges. The stationary construction does not by itself specify a universal post-collapse core, a prompt-collapse threshold, or the transient transport history that produced the final state.

10.3 Low-physical-mass and nonstellar compact objects

CPTG also permits black-hole-like compact states whose localized source mass is below ordinary stellar-remnant scales when the completed geometric response supplies part of the exterior gravitational charge. The admissibility question is geometric: whether the source and boundary data produce a trapped finite continuation with the appropriate GR exterior. Formation, stability, evaporation behavior, and any microscopic lower mass boundary are not determined by the stationary strong-field construction alone.

10.4 Supermassive compact objects

The same compact requirements apply to supermassive objects. Their finite continuation must preserve the measured exterior charges, satisfy the compact conservation relations, and recover the ordinary GR exterior wherever the additional response is negligible. The polarization–transport crossover remains object dependent rather than being assigned by total mass class alone.

10.5 Primordial black holes

CPTG permits primordial black holes formed before ordinary stellar evolution under the same finite compact solution class [10, 11]. They do not require a stellar progenitor. Their formation mechanism, mass function, survival, merger history, and abundance are cosmological quantities and require the native CPTG cosmological framework. Compact admissibility alone does not fix a primordial abundance or population fraction.

11 Strong-Field Scope and Claim Boundaries

The compact construction separates parent-theory requirements, properties of the registered finite realization, and observational readouts. Table 5 summarizes the authority and physical role of each element.

Table 5: Authority and role of the principal strong-field quantities used in this paper.

Element	Theory role	Physical role or boundary
Finite completed continuation	Parent strong-field requirement	The physical compact state remains finite rather than ending in an infinite singular continuation.
Kerr or Schwarzschild exterior	Retained GR branch	Controls the exterior wherever polarization and transport are negligible; ordinary horizons remain admissible.
Finite profile $F(x)$	Registered compact realization	Supplies the analytic interior used by the calculations in this paper and matches the constant-mass exterior through second radial derivative.
Matching radius r_m	Compact handoff boundary	Joins the registered finite interior to the retained exterior; it is not a universal response-activation radius.
Spherical source–transport–polarization mass state	Closed finite compact realization	Misner–Sharp state closes the physical-source, interior-transport, and geometric-polarization mass channels before rotation is introduced.
Conditional mass–spin handoff	Registered compact closure	Carries the closed spherical mass state to a rotating exterior benchmark while preserving M_∞ and total angular momentum.
Structural Mode N_{BH}	Kerr-compatible post-solution structural measurement	Measures the organization of the already solved compact state and fixes the organization shell r_N ; it is not a separate axisymmetric field solve.
Inner trapping surface r_-	Geometric readout	Distinct from both the organization shell and the exterior matching boundary.
Finite-core lapse and clock	Geometric readout	Measures local proper-time accumulation relative to source-local asymptotic exterior Kerr time.
ZAMO frame dragging	Geometric readout	Measures rotation of the local inertial frame; it is not a material or fluid angular velocity.
Formation history	Separate dynamical problem	Collapse, fallback, radiation, mergers, and primordial formation determine how a system reaches the stationary compact state.
Ringdown, tidal response, and secondary-wave structure	Perturbative observables	Not fixed by compact admissibility alone; they require perturbations of the solved background geometry.
Primordial abundance and survival	Cosmological population quantities	Not fixed by the stationary compact solution; they require native CPTG formation and population dynamics.
Present compact-sector scope	Declared calculation boundary	The paper treats the mass–spin compact sector. Electromagnetic charge is not included, and no neutron-star equation-of-state calculation is claimed.
Parent-action rotating uniqueness	Not established	The Kerr-compatible rotating readouts are downstream of the spherical closure and conditional handoff; they are not a uniqueness theorem or a completed general axisymmetric interior field solve.

Continued on next page

Element	Theory role	Physical role or boundary
Detector-level native inference	Not performed in the GWTC-5 calculation	Released observation-level products are translated into CPTG native coordinates before the compact solve; the strain likelihood itself is not refit.

12 Observational Consequences and Testable Regimes

Formation-sensitive systems constrain how exterior compact charges arise. Vanishing stars, failed-supernova candidates, lower-mass-gap binaries, and merger remnants are therefore useful when progenitor mass, ejecta or fallback, radiation, and angular momentum can be compared with the finite compact state. The compact mass decomposition must be applied to the measured exterior state rather than inferred from formation category alone.

Gravitational-wave observations test a different part of the construction. The GWTC-5 calculation in Sec. 8 tests repeated compact closure across posterior-resolved exterior states and separately propagates the official selection-corrected population posterior through the native CPTG coordinate. Ringdown, tidal response, or secondary-wave signatures are not universal consequences of the stationary compact equations; any such signal must be calculated by perturbing the solved finite background and matching the result to the retained exterior.

Population statements likewise depend on the population being studied. The GWTC-5 result is a push-forward of an existing selection-corrected binary-black-hole posterior, not a new formation-rate inference. Primordial or other nonstellar abundance calculations additionally require native CPTG formation histories, survival physics, merger evolution, and observational selection before comparison with lensing, gravitational-wave, accretion, or structure data.

13 Failure Conditions

The compact construction is falsifiable through its internal consistency and exterior-matching requirements. A physical CPTG compact solution must remain finite, respect the parent conservation structure, possess a well-defined polarization–transport response through the trapped region, and recover the corresponding GR exterior where the additional response becomes negligible. A compact state that necessarily enters an unbounded-curvature regime or cannot be joined consistently to its required exterior is not an admissible solution in that domain.

For the registered realization, the finite profile must satisfy the matching conditions of Eq. (6); the mass–spin handoff must preserve Eqs. (17) and (23); and the solved structural geometry must preserve the distinction among r_N , r_- , and r_m . An observational application fails its declared closure test if the native-coordinate reconstruction, compact cubic, mass decomposition, or conserved exterior identities do not close within the stated numerical tolerance.

Claims that require additional dynamics are tested at their own level. A proposed formation model must reproduce the relevant progenitor and remnant constraints, a perturbative waveform prediction must follow from the solved compact background, and a primordial population model must satisfy the observational limits applicable to its native CPTG abundance calculation.

14 Conclusion

CPTG describes compact objects with an ordinary General Relativistic exterior wherever curvature polarization and transport are negligible and a finite physical continuation in the deeper compact regime. Ordinary event and trapping horizons remain part of the solution. The registered finite realization joins to the retained exterior at $r_m = GM_\infty/c^2$ and separates the conserved exterior gravitational charge into a localized compact-source contribution and a completed geometric mass-equivalent contribution while preserving total angular momentum.

The same conditional mass–spin closure applies to the contrasting M33 X-7 and LMC X-3 benchmark states without object-specific compact coefficients. Their different exterior spin states produce different source/geometric decompositions through the same relation. The registered Kerr-compatible rotating readout then supplies Structural Mode N_{BH} , its organization shell, the inner trapping surface, local clock behavior, and geometric frame dragging as post-solution properties rather than fitting inputs.

The GWTC-5 validation extends that construction across 259 binary-black-hole events, 260 unique numerical streams, 520 compact components, and 7,626,239 retained posterior draws with no hard event rejection under the declared criteria. One fixed compact construction therefore remains closed across the declared catalog while resolving physical source mass, geometric mass contribution, Structural Mode, trapping structure, local clock behavior, and geometric frame dragging for the retained compact states. Structural Mode remains attached to each solved component, the ordering $x_N < x_- < 1$ is preserved across the posterior audit, and the finite-core clock and ZAMO frame-dragging operators remain downstream geometric readouts. The independent selection-corrected population calculation propagates all 8200 hyperposterior rows through the native-coordinate and compact-state maps, giving the median component-1 sequence $10.243 \rightarrow 9.623 \rightarrow 6.507 M_\odot$ from released source coordinate to CPTG exterior mass to physical compact-source mass.

The finite continuation also changes several standard black-hole questions without requiring an additional phenomenological ansatz. An infinite singularity is excluded as a physical endpoint, while the ordinary exterior horizon remains. The stationary Kerr or Schwarzschild exterior therefore continues to control exterior geodesic observables and supplies the background limit for ordinary ringdown and Hawking emission. The compact mass–spin relation predicts a nonextremal physical-source mass below the exterior gravitational mass, with the difference carried geometrically, and the internal solution fixes the ordering of the organization, trapping, and matching scales together with the limiting clock and frame-dragging behavior.

The authority of that result remains specific. The spherical Misner–Sharp three-channel

mass state is closed, the mass–spin map is a registered conditional handoff, and the Kerr-compatible Structural-Mode, trapping, clock, and frame-dragging quantities are downstream readouts of the already solved compact state rather than a completed general axisymmetric interior field solution. The GWTC analysis likewise begins from released parameter-estimation products rather than a native CPTG strain likelihood. These boundaries identify the next derivational and observational tests without changing the compact equations or the numerical closure demonstrated here.

Taken together, the calculations establish a finite compact state that joins an ordinary GR exterior while separating localized physical source mass from the larger conserved exterior gravitational charge whenever the state is nonextremal. The missing share is carried by the completed geometric response through the same source–transport–polarization organization that closes the spherical interior, while exterior spin fixes the conditional source/geometric partition without object-specific compact retuning. The same registered construction then assigns each solved object its own internal organization, trapping structure, local clock, and geometric frame dragging, and it remains operational from the contrasting stellar benchmarks through the full declared GWTC-5 catalog and population push-forward. In CPTG, the black hole is therefore not completed by an infinite singularity or an added invisible material component: it is a finite source–geometry system whose exterior remains General Relativistic and whose internal strong-field structure is calculable.

Data, Code, and Evidence Availability

The finite compact profile, matching conditions, conditional mass–spin handoff, stellar benchmark calculations, Structural-Mode construction, finite-core clock, and geometric frame-dragging operators are stated explicitly in this manuscript with their observational sources cited at the point of use. The catalog-scale GWTC-5 compact-state and population results summarized here are supported by the standalone publication archive

`CPTG_GWTC5_COMPACT_OBJECT_POPULATION_PUBLICATION_EVIDENCE_FINAL.zip`

The archive contains the ordered 259-event source registry and checksums, retained release-label identities and sample counts, event-level compact-state and Structural-Mode summaries, branch-resolved finite-core time and geometric-rotation tables and audits over 7,626,239 posterior draws, the frozen 8200-row native-coordinate population push-forward and coordinate diagnostic, implementation source, execution logs, numerical audits, and SHA-256 manifests [20]. Supporting CPTG framework papers and public research materials are maintained in the project repository [25, 26].

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